6. Transformers

Two coils sharing one field

Trading voltage for current

Lesson 6 · about 75 minutes · everyone builds; low-voltage AC only

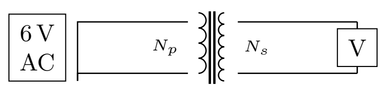
What you need

- A laminated iron core — a salvaged transformer core, or a bundle of iron wire
- Magnet wire, two colours if possible
- A low-voltage AC supply: a 6–12 V AC wall adapter (*AC output, not DC*)
- Multimeter with an AC volts range
- A small torch bulb and holder
- Tape, sandpaper

Predict first

You wind 100 turns on one side of a core and 200 on the other, and put 6 V AC into the 100-turn side. What comes out of the 200-turn side? And when you draw current from it, where does that energy come from?

Do it



laminated iron core

1. **Wind the primary.** 100 turns on one limb of the core, counted carefully. Sand the ends.
2. **Wind the secondary.** 200 turns on the other limb. Sand the ends.
3. **Step up.** Connect the AC adapter to the primary. Measure the AC voltage at the primary and at the secondary. Take the ratio.
4. **Step down.** Swap them — drive the 200-turn winding and measure the 100-turn one. The ratio inverts.
5. **Load it.** Put the torch bulb across the secondary. Measure the secondary voltage again — it sags. Now measure the *primary* current with and without the bulb connected.
6. **Try DC.** Swap the AC adapter for a battery of the same voltage. The output is zero, apart from a flick at the instant you connect or disconnect. Say out loud why, using Lesson 5.
7. **Remove the core.** Pull the two coils apart, or take out the iron. The output collapses.

What it means

The turns ratio $V_s/V_p = N_s/N_p$. Your measured ratio should be close to 2, and a little low because of losses.

Nothing is free $V_pI_p \approx V_sI_s$. Double the voltage and you halve the available current. A transformer trades one for the other; it does not create energy. The primary current rose when you loaded the secondary, which is that equation happening in front of you.

AC only A transformer needs a *changing* field. On DC it does nothing except at the moment of switching — which is Lesson 5 again, and it is also, exactly, what a spark gap will exploit.

The core's job The iron carries the field from one coil to the other so nearly all of it is shared. Without it, most of the primary's field misses the secondary. The fraction that is shared is called the **coupling**, and it becomes the central design question in Lesson 9.

Laminations The core is made of thin sheets, insulated from each other, precisely to break up the eddy currents you saw braking the magnet in Lesson 5. Here those currents are pure loss.

The link to the coil

The neon-sign transformer is exactly this device, with a huge turns ratio: about 120 V in and 9,000–15,000 V out, at correspondingly less current. It is also, by the same equation, why that supply is dangerous — the current it can deliver is small by industrial standards and still many times what stops a heart. The Tesla coil itself is *also* a transformer, but a strange one: no iron, loose coupling, and tuned to a single frequency. Lessons 8 and 9 explain why those three differences turn it into something a plain transformer can never be.

> **This lesson is low voltage.** Everything here runs from batteries or a small plug-in supply and cannot hurt you. That changes at Lesson 10. Build the habits now — one hand, discharge before touching, check before you power up — while the stakes are nothing, so that they are automatic when the stakes are real.